Thesis Defense Script

Full Script — All 29 Slides, Sections 1–6 — FINAL PRODUCTION VERSION

GOLD = say first — connecting words, transitions, bridge into the idea.
NAVY = main point / slide content.

Section 1 — Introduction

Slide 1 — Cover Page

Good morning, respected Chairman, Reviewers, Committee Members, and esteemed audience. My name is PHORN Sreypov, and today I'm honored to present my thesis, "Electrical Load Forecasting and Demand Management Using Meta-Stacking Ensemble Learning with a Smart Forecasting Platform." This research was conducted during my internship at the EDC–AFD–EU Platform for Research and Training on Power System, under the supervision of Mr. NOP Phorn Nara, with academic guidance from my advisor, Dr. HAS Sothea.

> *Speaker note: Memorize this slide closely — it's the only one read almost verbatim. Maintain direct eye contact with the committee, not the slide presentation.*

Slide 2 — Table of Contents

Before going further, Let me give you a brief overview of the journey we'll take today. I'll walk through six parts: the introduction to my internship and host organization, the project description and the problem that motivated this work, a review of related literature, the methodology I developed, the results and discussion, and finally the conclusion together with a short demonstration of the platform.

> *Speaker note: Say this fast and light — orientation, not content. Don't read the numbered bullets "01, 02, 03…" directly off the slide.*

Slide 3 — Internship Overview

Let's start with where this work took place. I completed a four-month hybrid internship, combining both on-site and remote work, at the Host Platform for Research and Training on Power Systems. I worked as a Data Scientist, focusing on electricity load forecasting and management, with a special emphasis on machine learning and deep learning models, using real-world operational power data from EDC.

Slide 4 — EDC-AFD-EU Project Overview

This internship is part of a larger project. Before I explain my specific work, let me give you a brief overview of the EDC-AFD-EU Project. It was created by EDC, funded by AFD, and is a collaboration between multiple partners. Our mission is to strengthen Cambodia's power sector by developing skilled professionals through education, research, and global collaboration at the Master’s and PhD levels.

Section 2 — Project Description

Slide 5 — Project Introduction

With that context in mind, I want to highlight the key concern that this project was designed to address. As you can see in this chart, Cambodia’s electricity demand is rising rapidly. Between 2010 and 2024, our power capacity grew almost nine times, and the energy delivered grew about eight times. The critical point is after 2019, when the growth became much steeper and less predictable. Official forecasts now show that demand will more than triple from today’s levels. This rapid, explosive growth is exactly why EDC can no longer rely on manual methods, and why we urgently need automated machine learning tools

> *Speaker note: This is a 'show the concern' slide — state the growth numbers with significant weight, as they form your core justification for urgency.*

Slide 6 — Problem Statement

Now that we've seen how quickly electricity demand is growing, the next question is: Can EDC's current forecasting process keep up with this growth?

Unfortunately, the answer is no. However, EDC still relies on manual forecasting. This makes it difficult to capture complex electricity demand patterns and creates **three major challenges**.

First, the data environment is inconsistent. Manual data cleaning is time-consuming and can introduce hidden errors, reducing forecasting accuracy.

Second, the forecasting workflow is not connected. Data preparation, model training, and prediction are handled separately, making the process slower and less efficient.

Third, there is no user-friendly platform. Operational staff do not have an easy way to generate automated forecasts quickly and efficiently.

These challenges clearly highlight the need for an integrated forecasting platform that combines machine learning with an easy-to-use web application.

Slide 7 — Project Objectives

To address these challenges, this research was designed with one main goal.

The primary goal is to develop a smart electricity load forecasting model and integrate it into an easy-to-use web platform for EDC.

To achieve this goal, the project has **four key objectives**.

First, identify the most important input features using different feature selection techniques.

Second, compare different feature sets to determine which one provides the best forecasting performance.

Third, develop and evaluate machine learning models using a rolling validation approach.

Finally, build an interactive web dashboard that allows non-technical users to generate forecasts easily.

Before moving on, I'd like to mention one important point. This system is designed only for **48-hour electricity load forecasting**. Real-time SCADA integration and automated grid control are **outside the scope of this research.**

> *Speaker note: Don't read the objectives like a dry list. Say "First... Second... Third... And fourth..." as a continuous sentence with your eyes on the committee.*

Slide 8 — Project Planning

Before going into the technical details, let me quickly show you how this work was actually planned and executed. Over a 16-week timeline, I ran parallel development workstreams. The ML Research Pipeline moved from data collection, feature engineering, and feature selection through model tuning, rolling validation, and stacking. In parallel, the Web Platform Development advanced through system architecture, backend implementation, frontend implementation, and production deployment. Our actual timeline tracked closely to the plan, serving as an efficient phase-gate development lifecycle.

Section 3 — Literature Review

Slide 9 — Literature Review

Before designing the methodology, it's worth grounding it in what's already been done in the field — so let's look at the literature. I benchmarked five key studies. Nara, Has, & Bun (2026) developed an hourly STLF model for the Cambodia EDC grid using a stacking meta-ensemble via linear regression, achieving a 3.86% MAPE, but left a gap with no automated feature selection and limited weather data. Yaprakdal & Arısoy (2023) introduced a multi-stage filter-wrapper feature pipeline yielding a 1.2% MAPE reduction, but omitted an ensemble stacking approach. Vianita & Tantyoko (2025) benchmarked distinct models on tropical climate datasets proving no single model dominates, but ran models in isolation. Taheri (2025) combined tree-based ensembles via a secondary meta-model but assumed predefined feature subsets. Finally, Beloev (2025) proved that accounting for operational telemetry delays prevents up to 15% artificial data leakage.

> *Speaker note: Use this spoken narrative as your summary while the table sits behind you as supporting technical detail. Do not read the table cell by cell.*

Section 4 — Methodology

Slide 10 — Overall Methodology

Now that we've discussed the objectives, let's move on to the methodology of this research. This methodology was designed to address the research gap identified in the literature. As shown in the diagram, the project consists of **two main stages**.

The first stage is the **Machine Learning Forecasting Pipeline**. It begins with collecting electricity load and weather data, followed by data preprocessing and feature engineering. Next, we apply feature selection, develop the forecasting models, and optimize them using rolling validation and hyperparameter tuning.

The second stage is **System Development**. In this stage, the optimized forecasting model is integrated into a web platform built with a **React** frontend, **FastAPI** backend, and **PostgreSQL** database.

Together, these two stages create a complete forecasting system, from data processing and model development to deployment in a practical web application.

Slide 11 — Data Collection

Let's start with the data, which forms the foundation of our work. We got the data from the EDC Grid Modernization Project. Our target is the total national grid load from 2019 to 2024. To improve on past work, we didn't just use one location. Instead, we added weather data from five locations—Phnom Penh, Sihanoukville, Siem Reap, Ratanakiri, and Svay Rieng—using Open-Meteo. This gave us a richer, multi-location weather set.

Slide 12 — Data Cleaning and Feature Engineering

After collecting the data, the next step was to prepare it for modeling.

First, we cleaned the data by detecting outliers and handling missing values to improve data quality.

Next, because the Open-Meteo historical data ended in **March 2021**, we combined historical weather data with forecast weather data to create one continuous dataset covering **2019 to 2024**.

Finally, we performed feature engineering by creating lag features, rolling statistics, calendar features, momentum features, and a working-day indicator. These new features help the model better understand electricity demand patterns and improve forecasting accuracy.

Slide 13 — Feature Selection Framework

After preparing the data, the next step was feature selection, which is the main contribution of this research.

Rather than relying on a single method, I developed a **three-stage feature selection framework** to identify the most informative features for electricity load forecasting.

In Stage 1, we applied statistical methods, including Pearson Correlation and Mutual Information, to select the initial candidate features.

In Stage 2, we evaluated feature importance using Permutation Importance and SHAP across three machine learning models: Random Forest, XGBoost, and MLP. This allowed us to identify features that were consistently important across different models.

Finally, in Stage 3, we compared different feature groups and selected the optimal feature set for the final **48-hour forecasting model.**

This three-stage framework helps improve feature selection and leads to more accurate and reliable forecasting results.

Speaker note: Speak with high confidence here — this is your unique academic contribution.

Slide 14 — Rolling-Year Validation - Rolling Validation, Model Development, and Hyperparameter Tuning

After selecting the optimal features, the next step was to develop and evaluate the forecasting models.

To ensure a fair evaluation and avoid temporal data leakage, we used a **4-fold expanding rolling-year validation**. As shown on the left, the training data gradually expands from **2019** to **2022**, while each following year is used for testing.

Importantly, the entire 2024 dataset was kept untouched throughout feature selection, model development, and hyperparameter tuning. It was used only once as the final holdout test to evaluate the completed model.

With the validation strategy in place, we trained **four base models**: Random Forest, XGBoost, SVR, and MLP.

Their out-of-fold predictions were then combined to train **two meta-learners**: Meta-Linear Regression and Meta-SVR, forming our **meta-stacking ensemble model**.

Finally, we optimized every model using **Grid Search**, selected the best configuration based on the **average MAPE across all four folds**, and evaluated the final model on the untouched **2024 holdout dataset**.

Slide 16 — Meta-Stacking Ensemble Learning Framework

Now I'd like to bring all the previous steps together and show how the complete meta-stacking framework works.

As shown in the diagram, the process begins with **data cleaning**, **feature engineering**, and **feature selection** to prepare the final dataset.

Next, we apply **rolling-year validation** and **hyperparameter tuning** to train the four base models: **Random Forest, XGBoost, SVR, and MLP**.

The predictions from these base models are then combined to create a **meta-stacked dataset**, which becomes the input for two meta-learners: **Linear Regression** and **SVR**.

Finally, the best meta-stacking model is evaluated on the untouched **2024 holdout dataset** using **MAPE** and **MAE**.

This framework combines multiple machine learning models into a single ensemble, producing a more robust and accurate forecasting model before deployment into the web platform.

Slide 17 — System Architecture and Development

After developing the forecasting model, the next step was to deploy it as a practical web platform for EDC.

The system is organized into **four main layers**.

The **Presentation Layer** provides a user-friendly interface for different types of users.

The **API Layer** handles communication between the web application and the forecasting model while ensuring secure access.

The **Machine Learning Layer** performs feature engineering and generates electricity load forecasts using the trained meta-stacking model.

Finally, the **Database Layer** stores user information, uploaded datasets, forecasting results, and system records.

The system also supports a complete automated workflow, from data upload and validation to forecasting and interactive result visualization.

Slide 18 — System Development and Quality Assurance

To ensure the platform is reliable and easy to maintain, the system was developed through four main stages.

These stages included system setup, machine learning integration, API development, and final testing.

The application was developed using **Docker** and an automated **CI/CD pipeline**, making deployment more efficient.

Finally, we verified the system through automated unit testing and end-to-end testing to ensure both the forecasting functions and the web application operate correctly.

As a result, the platform is ready to support practical electricity load forecasting for EDC users.

Section 5 — Results and Discussion

Slide 19 — Feature Selection Analysis (Baselines)

Now let's look at the experimental results, starting with the baseline feature selection methods.

The results show that **no single feature selection method worked best for every machine learning model.**

For example, **Pearson Correlation and Mutual Information** achieved the best performance for the **MLP** model, while **Permutation Importance** performed better for **Random Forest**. **SHAP** also favored tree-based models.

These results show that each model prefers a different feature subset.

This observation motivated our proposed Multi-Hybrid Feature Selection Framework, which combines the strengths of multiple methods instead of relying on only one.

Since no single method performed consistently well, we developed the proposed Multi-Hybrid Feature Selection Framework. Let me now show its results.

Slide 20 — Multi-Hybrid Feature Selection Results

Now let's look at the performance of the proposed Multi-Hybrid Feature Selection Framework.

The framework reduced the number of features from **66 to just 18**, representing a **72.73% reduction**.

Despite using far fewer features, the forecasting models maintained strong prediction accuracy.

In addition, reducing the feature space made model training and hyperparameter tuning much more efficient, significantly lowering the computational cost.

These results demonstrate that the proposed framework produces a smaller, more efficient feature set without sacrificing forecasting performance.

Slide 21 — Ensemble Model Performance (2024 Holdout)

Now let's evaluate the performance of the proposed meta-stacking ensemble on the untouched 2024 test dataset.

The results show that the **Meta-LR ensemble** achieved the best overall performance, with the **lowest MAPE of 3.61%** and the highest overall prediction accuracy.

More importantly, it outperformed all four individual base models, including Random Forest, XGBoost, SVR, and MLP.

Although we also tested **Meta-SVR** as a meta-learner, its performance was lower, indicating that **Meta-LR was better at combining the predictions from the base models.**

These results confirm that the proposed meta-stacking framework provides more accurate and reliable electricity load forecasting than any individual model.

Slide 22 — Monthly Forecast Performance - It is about showing **consistency**.

Next, let's examine how the model performs throughout the entire year.

As shown in the figure, the **Meta-LR predictions closely follow the actual electricity demand across all twelve months.**

The model successfully captures both the seasonal peaks and the lower-demand periods while maintaining stable forecasting accuracy.

This illustrates that the model is not only accurate overall but also remains consistent throughout different seasons of the year.

Slide 23 — Hourly Resolution Detail

Finally, let's look at the forecasting performance at the hourly level.

This figure shows one representative week from **March 2024**.

As we can see, the **Meta-LR model closely follows the actual electricity demand**, including both the daily peaks and overnight low-demand periods.

Compared with the individual base models, the meta-stacking model produces more stable and accurate predictions, especially during peak demand periods.

This demonstrates the practical value of the proposed ensemble model for real-world electricity load forecasting.

Slide 24 — Discussion

Now, I'd like to summarize the key findings of this research.

This study demonstrates **four main contributions**.

First, the proposed Multi-Hybrid Feature Selection Framework reduced the number of features from **66 to 18**, making the forecasting process more efficient while maintaining high accuracy.

Second, the feature importance analysis showed that historical load features, especially **lag-24** and **lag-168**, were the most influential predictors. Among the weather variables, **temperature** had the greatest impact on electricity demand.

Third, the proposed **Meta-Stacking Ensemble** achieved the best forecasting performance, with a **MAPE of 3.61%**, outperforming all individual machine learning models.

Finally, the forecasting model was successfully integrated into a smart web platform, providing an automated and user-friendly solution for electricity load forecasting at EDC.

Overall, these findings demonstrate that the proposed framework is both accurate and practical for real-world deployment.

Section 6 — Conclusion and Demonstration

Now that I've presented the methodology, results, and discussion, I'd like to conclude by reviewing how this research achieved its objectives.

Slide 25 — Outcomes

Let's return to the four objectives presented at the beginning of this defense.

I'm pleased to say that **all four objectives have been successfully achieved.**

First, the proposed feature selection framework successfully identified the most important input features using multiple feature selection techniques.

Second, our experiments confirmed that the **18-feature consensus set** provided the best forecasting performance.

Third, the proposed **Meta-LR stacking ensemble** achieved the best overall results, with a **MAPE of 3.61%** on the 2024 holdout dataset.

Finally, we successfully developed and deployed a smart forecasting platform, providing an integrated and user-friendly solution for EDC.

Overall, these outcomes demonstrate that the proposed framework is both accurate and practical for real-world electricity load forecasting.

Slide 26 — SDG Impact

Beyond its technical contributions, this research also creates a positive impact on sustainable development.

The primary contribution is to **Sustainable Development Goal 7: Affordable and Clean Energy**.

By providing accurate electricity load forecasts, the proposed system helps EDC improve generation planning, enhance grid reliability, and increase operational efficiency. These improvements support a more reliable and sustainable electricity supply for Cambodia.

In addition, this research also indirectly supports SDG 9 through digital innovation in the power sector, SDG 11 by improving the reliability of electricity services, and SDG 13 through more efficient energy management.

Overall, this research demonstrates how artificial intelligence can support both technological advancement and sustainable energy development.

Before concluding, I'd like to briefly reflect on the challenges I faced and the lessons I learned throughout this research.

Slide 27 — Challenges Overcome

Every research project comes with challenges, and this project was no exception.

Throughout this research, I overcame four major challenges.

First, selecting the best features was difficult because different feature selection methods produced different results.

Second, I had to combine data science research with full-stack system development, requiring both machine learning and software engineering skills.

Third, I managed the development of the forecasting model and the web platform in parallel to complete both within the project timeline.

Finally, the large number of hyperparameter combinations required significant computational resources and careful optimization.

Overcoming these challenges was not only important for completing the project, but also became a valuable learning experience.

> *Speaker note: State these challenges as professional engineering facts that you successfully managed. Keep your voice confident and objective.*

Slide 28 — Lessons Learned

Slide 28 – Lessons Learned

Completing this research has been a valuable learning experience, both technically and personally.

From a technical perspective, I strengthened my skills in machine learning, feature engineering, model optimization, and full-stack system development.

From a personal perspective, I improved my self-learning ability, problem-solving skills, project management, and communication.

Most importantly, this project taught me how to transform a research idea into a practical solution that can support real-world electricity forecasting.

Slide 29 — Future Direction

Although the proposed system achieved its research objectives, there are several opportunities for future improvement.

First, extend the forecasting horizon from **48 hours** to **168 hours** to support longer-term planning.

Second, explore advanced deep learning models, such as **LSTM, Bi-LSTM, and Transformer**, to further improve forecasting accuracy.

Third, implement automated model monitoring and retraining to maintain performance as new data becomes available.

Fourth, integrate the platform directly with **EDC's SCADA system** and weather APIs for fully automated data collection.

Finally, continue enhancing the web platform by adding new features and improving the overall user experience.

These future directions will make the system more intelligent, automated, and practical for real-world deployment.

Platform Demonstration – show what the user can do

Before concluding my presentation, I would like to give a brief demonstration of the Smart Load Forecasting Platform.

First, users log in to the system through a secure role-based interface.

Next, they can upload electricity load and weather data through the **Warehouse** page. The system automatically validates the uploaded data before processing.

Then, on the **Prediction** page, users simply select the forecasting period and run the **Meta-LR forecasting model**.

Within a few seconds, the platform displays interactive forecasting results, allowing users to compare the predicted and actual electricity demand. Users can also download the forecasting report for further analysis.

This demonstration shows how the proposed research has been transformed into a practical and user-friendly forecasting platform for EDC.

Closing Slide

This concludes my presentation.

Thank you very much for your time and attention.

I respectfully welcome any questions, comments, or suggestions from the respected Chair, committee members, and distinguished guests.